

Cody Appa, PhD

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cody-appa.github.io/Cody_Website/

Professional Summary:

Accomplished Molecular Biologist with 8+ years of research experience in advanced gene editing technologies including CRISPR interference. Recent PhD graduate with expertise in DNA synthesis and gene editing, mammalian cell culture, and bioinformatics. Published researcher with multiple peer-reviewed publications including an Editor's Pick first-author paper in mSphere. I am looking for a Research Scientist role in Biotech, Biopharma, or Clinical Biology.

Work Experience:

Research Assistant/PhD Candidate – University of Idaho:

August 2021 - August 2025

- Method Development and Novel Scientific Discovery: 8+ years of research experience; developing and driving molecular biology assays and DNA manipulation techniques. Culminated in discovering the Saturn Body, a transitional cell form in *Chlamydia trachomatis* development.
- Extensive Molecular Biology and Gene Editing Experience: Designed and executed complex DNA synthesis and cloning projects including designing, engineering, and cloning 100+ unique plasmids including CRISPR-cas systems and genetic biomarkers.
- Improved Plasmid Cloning Efficiency by over 90%: Improved efficiency and accuracy of genetic cloning SOP in a historically genetically intractable organism (*Chlamydia trachomatis*). Drastically reduced time of protocol from 3 hours to 45 minutes while increasing accuracy and reducing cross-over events.
- Cell Culture and Aseptic Technique: 8+ years of mammalian and bacterial cell culture experience in a Biosafety 2+ environment. Managed multiple concurrent cell lines with simultaneous infectious co-culture with zero cross contamination.
- Independent Project Leadership: Led multi-year research initiatives from experimental design through data analysis and publication, consistently meeting project milestones while managing multiple concurrent studies.
- Technical Documentation & Communication: Maintained good documentation practices and kept detailed experimental records, authored research publications, and presented findings to diverse scientific audiences.
- Quantitative Analysis & Bioinformatics Tools: Applied programming languages Python (matplotlib, NumPy, Pandas) and R (ggplot2) for experimental data analysis, sequence alignment, and statistical modeling, bridging wet lab work with data science.
- Mentorship & Teaching: Trained over 20 undergraduate researchers in advanced molecular biology techniques, demonstrating ability to share technical expertise and develop others.
- Cross-Functional Collaboration and Communication: Collaborated with cross-functional teams and departments. Multiple award-winning presentations and an Editor's Pick first author publication demonstrate ability to effectively communicate complex data.

Research Technician – University of Idaho: (Full-time post-baccalaureate research position)

August 2019 – May 2021

- Fluorescent Biomarkers and Transcriptomics: Created fluorescent promoter reporter plasmid constructs for 10+ genes of interest. Utilized fluorescent biomarkers for cell-type specific protein studies. Participated in transcriptomics including RNA-seq library prep and data analysis.
- Laboratory Operations and Maintenance: Maintained lab equipment, managed inventory and reagent preparation, ensuring continuous research operations.

Undergraduate Research Assistant – University of Idaho:

January 2017 – May 2019

- Research Experience: Developed E- riboswitch repression vectors. Transformed plasmids into *E. coli* and *Chlamydia*.
- Reagent Prep and Lab Maintenance: Created reagents, prepped media stocks, and maintained lab space.

Education:

PhD in Molecular Biology, Microbiology and Biochemistry
University of Idaho | August 2025 | 3.96 GPA

BS in Biology
University of Idaho | May 2019 | 3.2 GPA

Skills and Techniques:

DNA Manipulation/Gene editing

- PCR, qPCR, RT-PCR, ddPCR
- Plasmid design and plasmid biology, Ligation-based DNA assembly, and molecular cloning (Benchling, NCBI)
- CRISPR/Cas systems and plasmid design
- PCR primer design/troubleshooting
- Nucleic acid extraction, sequencing, and analysis
- DNA synthesis, assembly, scale-up, transformation, genome integration

Data Analysis/Visualization

- Proficient in Python (matplotlib/pandas) and R (ggplot2) programming languages for data analysis and visualization
- Familiar with AlphaFold Ai-guided protein design

Cell Culture

- Mammalian cell culture: Cos-7 cells and Hela cells
- Microbial Culture: E. coli, Yeast, Chlamydia/host co-culture
- Automated liquid handling systems for plate-based assays
- Maintained multiple cell lines simultaneously
- Fluorescent microscopy (Nikon and Andor systems)
- ELISA, multi-channel flow cytometry, FACS, and FlowJo analysis
- Fluorescent confocal microscopy, immunofluorescence, Fluorescent in-situ hybridization

Bioinformatics/NGS

- RNA-seq library prep, pooled screening, and multiplexing
- Reverse transcription, enzyme digestion, and droplet generation/analysis for ddPCR
- Protein isolation/expression assays including SDS/PAGE, chromatography, and spectrophotometry
- Bioinformatics data analysis via Python

Honors:

University of Idaho College of Science Student Research Expo

Best Graduate Student Poster
-March 2025

Editor's Pick First Author Publication

<https://doi.org/10.1128/msphere.00437-24>
-August 2024

Max and Sharon Walker College of Science Scholarship

-November 2024

Chlamydia Basic Research Society Bi-annual International Conference

Jane Raulston Award for Best Graduate Student Poster
-March 2023

Publications:

Nicole A. Grieshaber, **Cody Appa**, Megan Ward, Alorah Grossman, Sean McCormick, Brendan S. Grieshaber, Travis Chiarelli, Hong Yang, Anders Omsland, Scott S. Grieshaber. The T3SS structural and effector genes of *Chlamydia trachomatis* are expressed in distinct phenotypic cell forms.

doi.org/10.1101/2024.04.25.591156

May 2025

Appa CR, Grieshaber NA, Yang H, Omsland A, McCormick S, Chiarelli TJ, Grieshaber SS. The chlamydial transcriptional regulator Euo is a key switch in cell form developmental progression but is not involved in the committed step to the formation of the infectious form.

doi.org/10.1128/msphere.00437-24

August 2024

Chiarelli TJ, Grieshaber NA, **Appa C**, Grieshaber SS. 2023. Computational modeling of the chlamydial developmental cycle reveals a potential role for asymmetric division.

doi.org/10.1128/msystems.00053-23

March 2023

Nicole A. Grieshaber, Travis J. Chiarelli, **Cody R. Appa**, Grace Neiswanger, Kristina Peretti, Scott Grieshaber. Translational gene expression control in *Chlamydia trachomatis*.

doi.org/10.1371/journal.pone.0257259

January 2022

Nicole A. Grieshaber, Justin Runac, Sierra Turner, Marissa Dean, **Cody Appa**, Anders Omsland, Scott Grieshaber. The sRNA regulated protein DdbA is involved in development and maintenance of the *Chlamydia trachomatis* EB cell form.

doi.org/10.3389/fcimb.2021.692224

July 2021